AI & ML Lab – Final Lab Assignment Links

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CNN Image Classification – Geometric Shapes

Repository and Notebooks

1. **GitHub Profile:**
   https://github.com/ahsanjust/

2. **Final Lab Standalone Assignment Repository:**
   https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification

3. **CNN – GitHub Notebook:**
   https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/220119_CNN.ipynb

4. **Google Colab Notebook (GitHub-Native, Direct Link):**
   https://colab.research.google.com/github/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/220119_CNN.ipynb

5. **Repository README:**
   https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/readme.md

Performance Summary – CNN (Standard Test Set)

Class	Precision	Recall	F1-Score	Support
Circles	0.9643	0.9000	0.9310	30
Squares	1.0000	0.9667	0.9831	30
Triangles	0.9091	1.0000	0.9524	30
Overall Accuracy		**0.9556 (95.56%)**		

Architecture & Workflow Summary

- **Model:** 2-Layer CNN (PyTorch) – Conv2d(3→32) → ReLU → MaxPool2d(2) → Conv2d(32→64) → ReLU → MaxPool2d(2) → Flatten → Linear(16384→128) → ReLU → Linear(128→3)
- **Input Preprocessing:** 64×64 RGB images with single shared transform: Resize → ToTensor → Normalize($\mu = 0.5, \sigma = 0.5$)
- **Training Setup:** CrossEntropyLoss, Adam optimizer (lr = 0.001), batch size 64, 10 epochs
- **Dataset:** Geometric Shapes (Option 5) – train 543, validation 180, test 90, plus 10 custom smartphone photos

- **Real-World Accuracy:** 9 / 10 (90.0%) on custom smartphone photos (Circles: 4/4, Squares: 2/3, Triangles: 3/3)
- **Zero-Manual-Upload Automation:** Clones repo automatically in Google Colab, loads weights from `model/220119.pth`, evaluates test set, and predicts on custom photos automatically upon clicking "Run All".